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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Backtracking

b. Dynamic programming

c. Divide and conquer

d. Greedy method

The correct answer is:
Divide and conquer

Question **2**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()

b. sort()

c. sorted()

d. order()

The correct answer is: sorted()

Question **3**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It makes the search operation slower
- c. It has no effect on the search operation
- d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **4**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Being the most efficient sorting algorithm
- b. Sorting data in a non-sequential manner
- c. Bubbling up the largest element to its correct position with each pass
- d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **5**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is less efficient than bubble sort
- b. It does not require recursion
- c. It sorts data using a single pass
- d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Removing duplicates from a list
- c. Finding a specific element in a list
- d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **7**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Rearranging
- c. Merging
- d. Searching

The correct answer is: Sorting

Question **8**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the smallest element
- b. Sorting the entire list in one pass
- c. Bubbling up the largest element to its correct position
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Binary Search
- b. Linear Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Linear Search
- c. Binary Search
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **11**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Sorting
- b. Complexity
- c. Searching
- d. Merging

The correct answer is:
Complexity

Question **12**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. sort()
- c. order()
- d. arrange()

The correct answer is: sort()

Question **13**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It has no effect on the analysis process
- b. It complicates the analysis process
- c. It allows for quicker identification of duplicates
- d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Decreases the efficiency of algorithms
- b. Makes code execution slower
- c. Makes data harder to manage
- d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Sorting the entire list sequentially
c. Dividing the input into smaller subproblems
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems